Aegis - Technology Glossary

A plain-English decoder for the names in the Aegis setup. The reassuring point: almost none of this is exotic - Ed25519, namespaces, Landlock, seccomp, Z3, Cedar are standard, battle-tested components. We did not invent cryptography or sandboxing. The two original pieces are the **query compiler** (the AI fills a form, we write the safe query) and wiring **information-flow control** onto a kdb+ agent. Everything else is well-understood parts assembled into one fail-closed, signed, audited gate.

What we're protecting

kdb+	The database itself - a very fast database banks use for huge volumes of market data and trades.
q	The language you talk to kdb+ in. Extremely powerful (a full programming language) - which is the whole problem: the same language that asks ‘average price’ can also delete data or run OS commands.
HDB	‘Historical database’ - the on-disk pile of past data, split by date.

The gatekeeper (the core gate)

PDP (Policy Decision Point)	Jargon for ‘the gatekeeper’: the separate program that says yes/no to every action.
Query compiler	Takes the AI’s structured *form* and writes the safe q itself. The ‘AI never writes code’ piece. **(The genuinely novel bit.)**
Detector packs	Pattern scanners (secrets, personal data, destructive commands) that act as a backup veto.
Egress proxy	‘Egress’ = outbound network traffic. A checkpoint all outbound traffic must pass; only approved destinations are allowed and the payload is inspected. Stops data being smuggled out.

The AI-specific defences

IFC (Information-Flow Control)	The ‘trusted vs untrusted’ tagging that stops prompt injection. Tracks where each piece of data *came from* and blocks untrusted data from triggering a dangerous action.
Row-level entitlements	Per-person row filters - which rows each user may see, stapled onto every query, non-removable.
RBAC	‘Role-Based Access Control’ - standard ‘permissions by job role.’ Off-the-shelf concept.

Making the rulebook un-forgeable + keeping records

Ed25519	A standard digital-signature algorithm. Signs the rulebook so the AI can’t forge or edit it - like a wax seal only the control function can make and anyone can verify.
WORM (Write Once Read Many)	A log you can add to but never edit or delete. The tamper-evident audit trail.
Hash-chain	Each log entry carries a fingerprint of the one before it, so quietly altering or deleting an entry breaks the chain and is detectable.

The cage (operating-system confinement - all standard Linux features)

Namespaces	A Linux feature giving the AI its own private view of the machine (its own files, network, process list) so it can’t see or touch anything else. The cell walls.
Landlock	A Linux security feature that lets a program lock itself to a specific allowed set of files, enforced by the kernel and irreversible. Limits which files the AI can reach.
seccomp	A Linux feature that filters which low-level system commands (‘syscalls’) a program may make. We block the dangerous ones (loading kernel code, debugging other processes, mounting disks). Shrinks the attack surface.
rlimits / no-new-privs	Resource caps (can’t fork-bomb the box) and ‘can never gain more privileges than it started with.’

The watchdog

Supervisor / circuit-breaker / kill switch	Watches the *stream* of actions and automatically quarantines and shuts the AI down if it starts misbehaving.
Overseer	A *second* AI that reads the logs and writes plain-English incident summaries. **Advisory only** - it never makes the security decision (you never want one AI guarding another as the real control).

Proving it's correct (the assurance layer)

Z3 / SMT	A mathematical theorem prover. Used to *prove* the rule-logic is sound (e.g. ‘an allowed action is always within granted permissions’), not just test it.

Cedar	Amazon’s open-source permissions language + checker. Used as an independent second opinion on our rules.
AgentDojo	The industry-standard security exam for AI agents - a large set of attack scenarios researchers score systems against.
MCP (Model Context Protocol)	The emerging standard way AI agents plug into external tools. We have per-tool rules for it.

The one line: “We didn’t invent crypto or sandboxing - those are standard parts. The original work is the query compiler and the information-flow defence, assembled with the rest into one fail-closed, signed, audited gate around the AI.”